

The Barbelo Prime Tree II

Visualization of OEIS Sequence A333489

(Anti-Run Compositions)

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ABSTRACT

This work presents an interactive browser visualization overlaying the integer sequence A333489 (OEIS) onto the Barbelo Prime Tree. The underlying prime tree was introduced in *Primzahlen Baum I* (Barbelo & Claude, 2026, DOI: 10.5281/zenodo.20475174): prime numbers form a vertical trunk while composite numbers between consecutive primes branch alternately to the left and right — analogous to a fishbone structure. This geometry makes the distribution of prime gaps directly visible.

The OEIS sequence **A333489** contains all natural numbers k for which the k -th composition in standard order is an *anti-run* — meaning no two adjacent parts are equal. The composition is derived directly from the binary representation of k : the leading 1 begins the first part; each subsequent 1 starts a new part, and each 0 increments the current part by 1.

A333489: 0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 16, 17, 18, 20, 22, 24, 25, 32, 33, 34, 37, 38, 40, 41, 44, 45, 48, 49, 50, 52, 54, 64, 65, 66, 68, ...

In this visualization, all numbers belonging to sequence A333489 are additionally highlighted in **magenta** within the prime tree — regardless of whether they are prime (trunk) or composite (branch). This allows the distribution pattern of anti-run compositions to be read directly from the geometric tree structure and related to the distribution of prime numbers.

The interactive interface supports zoom, pan, search, and a range slider from 2 to 1,000. The implementation uses pure HTML5 and Canvas with no external libraries. To the best of the authors' knowledge, no comparable visualization of this sequence in a prime tree context exists.

KEYWORDS

Keywords: prime numbers · prime tree · OEIS A333489 · anti-run compositions · compositions in standard order · integer sequences · interactive visualization · HTML5 Canvas · number theory

RELATED WORK

This work is a supplement to *Primzahlen Baum I* (is supplement to):
doi.org/10.5281/zenodo.20475174

REFERENCES

- Barbelo, N.J. & Claude (Anthropic). (2026). *Primzahlen Baum I: A New Visual Representation of Prime Gap Structure*. Zenodo. <https://doi.org/10.5281/zenodo.20475174>
- Sloane, N.J.A. et al. *The On-Line Encyclopedia of Integer Sequences*. Sequence A333489 — Numbers k such that the k -th composition in standard order is an anti-run. <https://oeis.org/A333489>